

Q: Vision Transformer for High-Resolution Mango Leaf Disease Detection

You are provided with a high-resolution mango leaf disease dataset (image size: 2000×1000), consisting of images labeled into multiple disease categories (e.g., *Healthy*, *Anthracnose*, *Powdery Mildew*, *Sooty Mold*, etc.).

Tasks:

1. Preprocessing:

- Resize each image while preserving its aspect ratio to 224×112.
- Pad the resized image to a square size of 224×224 (center padding).
- Normalize the pixel values using mean = [0.5, 0.5, 0.5] and std = [0.5, 0.5, 0.5].
- Split the dataset into training (80%) and validation (20%) sets.

2. Model Preparation:

- Load a pre-trained Vision Transformer (vit_base_patch16_224) using either timm or transformers.
- Modify the classification head to match the number of classes in the mango leaf dataset.
- Train (fine-tune) the model on your dataset for at least 5 epochs using GPU (if available).

3. Evaluation:

- Report training and validation accuracy after each epoch.
- Generate a confusion matrix and classification report (precision, recall, F1-score).
- Display 10 validation images with their predicted and ground truth labels.

4. Attention Visualization:

- Choose one image from the validation set that is correctly classified.
- Extract the attention weights from any one transformer layer.
- Visualize the attention maps (using matplotlib or seaborn) and interpret where the ViT model is focusing to make the prediction.

5. Report Submission:

Prepare a short report that includes:

- Screenshots of key results (sample predictions, attention visualization).
- Interpretation of the attention maps in your own words.
- Discussion: How does attention differ from CNN feature maps? What are the challenges in classifying high-resolution agricultural images?